Deep Learning Laboratory – Assignment 6

Team 17: Mirthula M (26120121), Karthikeyan M (26120004)

1. Sentiment classification

Fine-tuned GloVe 50d embeddings feed one vanilla tanh RNN with 64 hidden units. Concatenated temporal mean/max pooling feeds a two-class classifier. Training uses class-weighted cross-entropy and AdamW; validation macro F1 selects checkpoint 9 from 17 epochs. The prediction threshold is fixed at 0.5.

Of 5,600 supplied training rows, 399 normalized-text test overlaps and 526 remaining duplicate text/label rows are removed before grouped validation. Fit/validation/test sizes are 3740/935/1400; labels and the supplied test split are unchanged.

Score (%)	Train (fit)	Test
RNN accuracy	95.13	93.57
Majority baseline (always Negative)	93.13	92.50
RNN balanced accuracy	80.99	72.02

The test set has 1,295 Negative and 105 Positive labels: high accuracy alone is misleading. Test positive recall is 46.67%; duplicate test texts remain (1,340 unique texts in 1,400 rows), so rows are not independent examples.

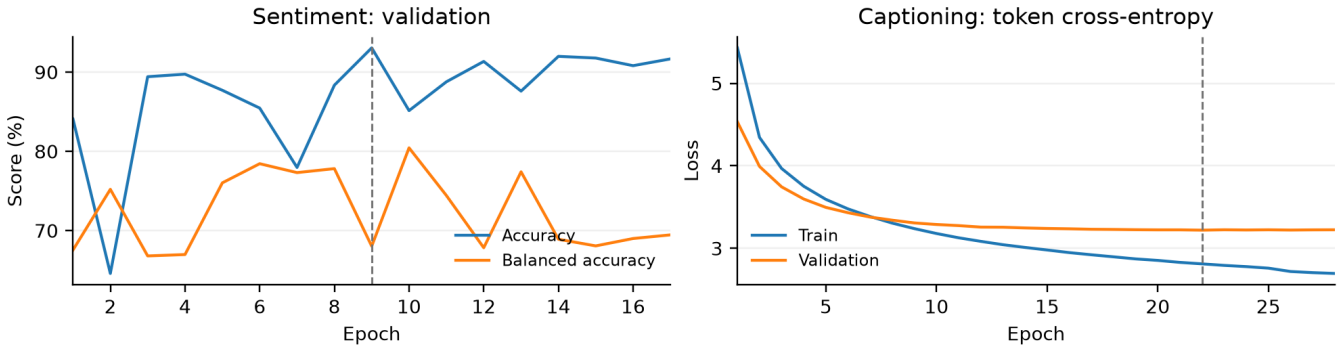
2. Image captioning

Frozen ImageNet VGG16 features are globally averaged to 512 dimensions, standardized using training statistics, and projected to a 64d context. Each GloVe 50d word vector is concatenated with this context and passed to one vanilla RNN (64 units), also initialized by the context. Embeddings are fine-tuned; Adam minimizes token cross-entropy. Best validation loss selects epoch 22 of 28.

The seed-17 image-disjoint train/validation/test split is 3200/400/400, with five captions per image. Vocabulary and feature statistics use training data only.

Split	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Train	0.6626	0.4970	0.3557	0.2477
Test	0.5284	0.3494	0.2244	0.1425

Scores are unsmoothed cumulative corpus BLEU (0–1), over the full train/test splits with all five references. Evaluation is free-running beam search (beam 3, length penalty $\alpha = 0.7$, maximum sequence length 30), not teacher forcing. The lower test BLEU indicates weaker caption generalization.



Dashed lines mark selected checkpoints.

Pretrained: Stanford GloVe 6B (50d); torchvision VGG16 ImageNet-1K V1. The following galleries are random, not ranked; generated captions are unedited.

3. Training image examples

Five images sampled with seed 17 from sorted filenames; first two saved references shown.



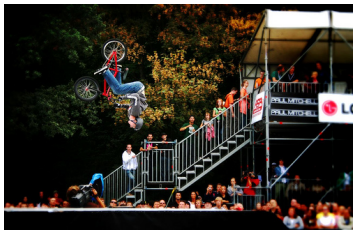
1. 3413806271_17b7e102aa.jpg

Reference 1: the people in the red parasail are gliding over the water
Reference 2: two people are hang gliding
Generated: two people are in the water



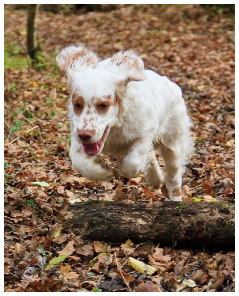
2. 317641829_ab2607a6c0.jpg

Reference 1: a man in a red coat stands outside of a store near a big column
Reference 2: a man in a red jacket and jeans stands next to a column in front of a store
Generated: a man in a red jacket is standing on a sidewalk



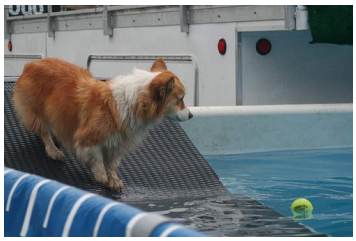
3. 2865564810_5c63328cd4.jpg

Reference 1: a biker doing a somersault with the bike in front of a crowd
Reference 2: biker performing jump while onlookers watch
Generated: a man is doing a trick on a ramp



4. 3048461682_e89f81b1c7.jpg

Reference 1: a dog runs over a stick in the leaves
Reference 2: a small white dog is jumping over a branch on the ground covered with leaves
Generated: a white dog is running through the grass



5. 2822148499_eaa46c99d4.jpg

Reference 1: a brown and white dog is looking at a tennis ball floating in a pool
Reference 2: a brown and white dog looking into a pool at a tennis ball
Generated: a white dog is jumping over a tennis ball

4. Test image examples

Five images sampled with seed 17 from sorted filenames; first two saved references shown.



1. 3350260112_fcb47ff6b2.jpg

Reference 1: lots of people are playing in the river that runs through the city and in the sandy park beside it
Reference 2: lots of people play in the water at the beach
Generated: a group of people are in the water



2. 3173014908_b3e69594b6.jpg

Reference 1: a person skiing downhill in the snow
Reference 2: a skier puffs up a cloud of snow
Generated: a skier is going down a snowy hill



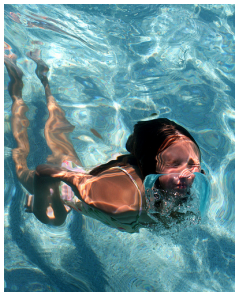
3. 2842609837_b3a0b383f7.jpg

Reference 1: a group of four friends are drinking alcohol at a party
Reference 2: a man wearing black stands with three woman who have blonde hair
Generated: two men are posing for a picture



4. 3041487045_b48ac7ed08.jpg

Reference 1: a dog runs through the woods
Reference 2: a pale tan dog romps through a wooded area
Generated: a dog runs through the snow



5. 2782480767_064c95eff2.jpg

Reference 1: a girl holds her breath underwater in a pool
Reference 2: a girl is about to emerge from a swimming pool
Generated: a little girl in a blue shirt is sliding down a water slide